

Mitigating Drought-Related Financial Risks for Water Utilities via Integration of Risk Pooling and Reinsurance

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Abstract: Concerns over drought-related financial risk have been growing for water utilities as adaptive measures, such as conservation (which reduces revenues) and water transfers (which increase costs), play larger roles in drought management. Recent research suggests that both utility-specific and generalized forms of index insurance can play an effective role in managing a water utility's drought-related financial risk. However, any entity offering these contracts to the water utility sector, whether a third-party insurer or a "mutual" formed by multiple utilities, will need to manage the "tail" risks of severe or widespread drought impacting many utilities simultaneously, which has the potential to drive high aggregate financial losses. When managing these risks, this entity must balance the opportunity costs of capital associated with higher reserve funds against the costs of mechanisms for transferring this tail risk (e.g., reinsurance). This research characterizes the financial risks of drought for a mutual, composed of surface water utilities distributed across all 344 climate divisions of the contiguous United States, that insures utilities against drought-related financial losses using contracts based on the Palmer Hydrologic Drought Index (PHDI). Tradeoffs associated with the mutual's risk management strategy involve varying levels of reserves and reinsurance. Results show that risk pooling via a mutual combined with reinsurance coverage can reduce the average annual net cost of risk management for member utilities by 7.5% while significantly reducing the average total net cost over 10-year periods by 17%. DOI: [10.1061/\(ASCE\)WR.1943-5452.0001202](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001202). © 2020 American Society of Civil Engineers.

Introduction

Reliance on supply-side solutions for managing drought, typically through the maintenance of large volumes of rarely used capacity, has become less practical due to the higher costs and limited availability of new sources and more restrictive environmental permitting (NRC 2012; Brajer et al. 1989; Leurig 2010; Hossain et al. 2019; Easter and Renwick 2018; Dilling et al. 2019). Consequently, more adaptive solutions, such as conservation or transfers of existing supplies, have become increasingly attractive (Lund and Israel 1995; Beecher and Chesnutt 2012; Gleick 2003; Ho et al. 2017; Kundzewicz et al. 2007; Brown and Carriquiry 2007; Gorelick et al. 2018). These approaches can, however, result in significant financial losses during drought because conservation often reduces revenues from decreases in water sales, whereas temporary supply augmentation (e.g., transfers) can increase costs because these are not included in annual budgets (Hughes et al. 2014; Tiger et al. 2014). The intermittent nature and varying levels of these losses correspond to drought frequency and severity, both of which are uncertain and add to the challenges of managing the accompanying financial risk (Olmstead and Stavins 2009; Donnelly and Christian-Smith 2013; Schmidt and Lewis 2017). Because water utilities typically employ a straightforward budget

model in which prices are set to recover costs and often have difficulty raising prices quickly in response to drought, even a 5% reduction in revenues can pose serious financial challenges (Zeff et al. 2014). Financial disruptions can lead to downgrades of a utility's credit rating, and the water utility sector's greater reliance on adaptive measures has begun to manifest in increased financial instability for many utilities (Chapman and Breeding 2016). As a result, ratings agencies have introduced new evaluation criteria designed to account for the effects of hydrologic variability, and a rating downgrade will lead to higher interest rates and increased debt service costs, jeopardizing a utility's long-term financial health (Chapman 2012; Donnelly and Christian-Smith 2013).

Managing these financial risks is typically accomplished through some combination of a reserve fund and drought pricing (Leurig 2012; Hughes and Leurig 2013). However, previous research has evaluated these tools in terms of their advantages and limitations and demonstrated that financial instruments, such as index-insurance, can be comparatively more effective in managing the risks posed by more severe drought (Zeff and Characklis 2013; Zeff et al. 2014; Baum et al. 2018). The literature has described financial instruments that have been tailored to individual water utilities (Zeff and Characklis 2013; Brown and Carriquiry 2007) as well as standardized instruments that have been applied to a geographically distributed pool of water utilities (Baum et al. 2018). Pooling financial risks over broad geographic regions involving (mostly) uncorrelated drought risks can lower the insurer's capital reserve requirements, often set by regulators at very high levels, up to the 99.5% value at risk (VaR) level (i.e., a level that ensures the ability to provide payouts with 99.5% confidence) (Johnson 2015; IAIS 2017). Risk pooling allows a reduction in the size of the reserves, lowering the costs associated with raising or maintaining reserves and thereby leading to the potential for reduced premiums (Baum et al. 2018). Lower costs may lead to greater uptake of these financial tools by utilities and the development of less expensive financial risk management strategies overall.

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Even with risk pooling, however, severe (low-probability, high-risk) events can drive insurance payouts to levels that require the maintenance of very large reserves (Johnson 2015; IAIS 2017). Because reserves must be maintained in a liquid form in order to be rapidly accessed, they typically earn a very low return, giving rise to high opportunity costs of capital. One way to reduce these costs is for an insurer to transfer some portion of the risk to a third party, often via reinsurance (Edesses 2015), reducing the required reserves. Reinsurance is often used by insurers to reduce tail risk associated with very rare, but costly, events (Schanz et al. 2010; Scordis and Steinorth 2012), effectively providing insurance to an insurer (Fig. 1).

This analysis is undertaken from the perspective of a mutual, which operates in much the same way as a risk pooling insurer, except that it is collectively owned by its members. A mutual typically pools similar, but independent, risks, each of which is relatively small compared to the pool itself, as a means of reducing variability in total payouts (Smith and Kane 1994), with auto collision insurance providing an illustrative example. Reinsurers, however, typically have very large capital pools and make use of diversification to manage a more finite set of large, uncorrelated risks, reducing the probability of larger-than-expected payouts, for example, by providing coverage against losses from natural disasters in multiple locations (Wu 2015). In this way, reinsurers often have a lower opportunity cost of capital relative to a mutual with a more singular exposure, which thereby allows the reinsurer to provide a cost-effective alternative to a mutual seeking to reduce its exposure to the risks of extreme events (Subramanian and Wang 2018).

When considering reinsurance, an important question arises as to the amount of risk a mutual should transfer to a reinsurer. One presumed objective, in this case, would be to minimize the costs of meeting regulatory reserve requirements designed to assure that payouts can be made. Although purchasing reinsurance reduces the capital costs for an insurer, reinsurance also involves costs, and these vary based on the probability and magnitude of the covered risk (Cummins et al. 2009; Froot 2001). Thus, insurers often find it practical to purchase reinsurance when the benefits of reducing the reserve requirement (i.e., lower costs of capital) outweigh its costs (Fu and Khury 2010).

Previous work has assessed the benefits of pooling drought-related financial risks for a set of spatially distributed water utilities (Baum et al. 2018), but long tail risks associated with widespread or severe drought still impose large reserve requirements on an insurer or mutual. However, the degree to which a mutual might cost effectively transfer some of this long tail risk to another party, namely through reinsurance, has not been explored. Nonetheless,

studies from other sectors with weather-related risks, such as agriculture, have investigated the joint use of reinsurance and risk pooling to mitigate financial losses from adverse weather (Porth et al. 2015, 2016). These studies used historical records of insurance, reinsurance, and loss data to identify combinations of risk pooling and reinsurance that significantly outperformed pooling alone, resulting in lower costs and a lower risk of insolvency for the insurer. Although weather is often a dominant risk, most crop insurance is a form of indemnity in which payouts are based directly on realized losses. This work focuses on index insurance, in which losses are tied to a specified environmental metric, rather than an assessment of the losses themselves. Index-based instruments can have some distinct advantages (e.g., lower transaction costs, less subjectivity in determining payouts, less concern over moral hazard), but their effectiveness is linked to the degree of correlation between the index and the realized losses.

This research is conducted from the standpoint of a mutual that must choose the level of reinsurance that will minimize its costs of meeting its regulatory reserve requirements (as measured by the 99.5% VaR). The mutual has pooled drought-related financial risks for a spatially distributed group of water utilities via index insurance contracts linked to the Palmer Hydrologic Drought Index (PHDI), an index chosen based on earlier work that compared its effectiveness with several other candidate indices (e.g., precipitation, streamflow) (Baum et al. 2018). The analysis is conducted for a generalized group of publicly owned surface water utilities distributed across all 344 climate divisions of the contiguous United States. The first objective is characterizing the distribution of financial losses from drought events of varying severity and spatial extent over the 120-year historical record of PHDI values observed in each climate division. Then, the financial losses for each climate division are estimated and pooled for analysis over a synthetic 100,000-year record based on the 120-year historical record. The net cost (a more explicit definition will be provided later) of insuring utilities with various reinsurance contract designs is then evaluated, and the reinsurance contract resulting in the lowest average annual net cost is used to determine the net cost of a combined pooling and reinsurance strategy over a 10-year period. Results should provide insight into potential approaches for better managing drought-related financial risks over individual and multiyear periods.

Methods

A key consideration in developing a mutual, as opposed to engaging with a third-party insurer, is that a mutual is owned by its

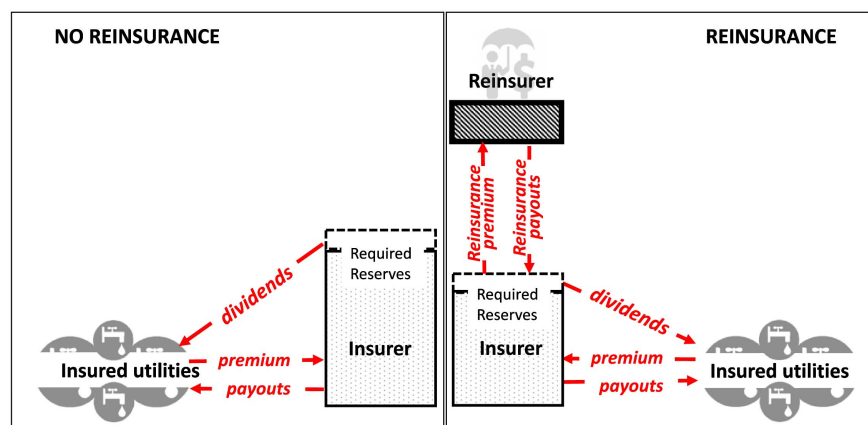


Fig. 1. Flow of funds for insuring with and without reinsurance.

policyholders, who contribute capital to build up the mutual's reserves. As a result of contributing their own capital, mutuals will typically charge lower premiums (Pain et al. 2016; AMICE 2016; Hartwig and Alldredge 2018). Premiums are a function of both the insurer's estimate of expected payouts and the additional "loading" on the contract, with the latter accounting for the insurer's administrative costs, product development, and opportunity costs of capital associated with maintaining regulatory reserve requirements (the 99.5% VaR). Although a mutual can build its reserves slowly over time through assessments on its members in the early years after its formation, a newly formed mutual will establish the appropriate reserve level immediately by issuing debt, typically in the form of a bond. This approach allows it to meet specified reserve requirements without a long waiting period and then assess members the cost of servicing this debt over time. In this work, these costs are referred to as "debt service costs." The distribution of payouts to member utilities is determined by assessing the frequency and magnitude of payouts to a subset of insured surface water utilities from across the United States. The payouts come as a result of these utilities using a PHDI-based index insurance contract to limit financial losses arising from severe drought events. Payouts are estimated based on nationwide simulations of PHDI, developed using an approach that maintains spatial and temporal autocorrelation across all 344 climate divisions in the contiguous United States. The generalized structure of index insurance contracts based on PHDI is described, with contracts tailored to each climate division by extrapolating results from earlier research based on financial data and drought metrics obtained from 372 surface water utilities distributed across the United States (Baum et al. 2018).

Once the distribution of aggregate annual payouts made by the mutual is determined, an analysis is conducted to evaluate the mutual's lowest-cost strategy for meeting a given certainty level in the ability to make these payouts. This strategy involves maintaining its own reserves through a combination of debt issuance (i.e., bond) and member premiums (both of which have costs to members) and transferring some portion of the "tail risk" of more extreme events to a third party via reinsurance (which also has a cost, primarily that associated with the loading charged by the reinsurer). Reinsurance contracts are considered across a range of thresholds to identify the lowest-cost combination of risk pooling and reinsurance, evaluating the costs realized by the mutual's member utilities in any single year, as well as over a 10-year period, and tracking the stability of the mutual's reserve levels over multiple years.

Generating Synthetic PHDI Data

Earlier work described the correlation between utility revenues and a range of drought-related metrics (Baum et al. 2018). These included the PHDI and cumulative measures of streamflow and precipitation (e.g., annual, seasonal, monthly) using data from 372 publicly owned, geographically diverse surface water utilities spread across 78 different climate divisions in the United States. The PHDI is a measure of drought, normalized to conditions in each climate division, with possible values between -12 and 12 . Positive values indicate wetter conditions, whereas negative values indicate drier conditions, with values above an absolute value of 3 representing severe conditions. Thus, a PHDI of -4 indicates very severe drought. It is also important to note that a similar PHDI level measured in two different climate divisions represents the same relative deviation from historic norms, not identical drought conditions, given that some regions are wetter or drier than others.

From a financial perspective, utility revenues are the primary metric of interest because utility costs are typically fixed and

mostly a function of debt service (Hanemann 1997; Hughes et al. 2014), making variations in revenue a reasonable surrogate for net revenues, the primary measure of financial stability (Zeff and Characklis 2013; Zeff et al. 2014). Annual revenue data from Moody's Water and Sewer Municipal Financial Ratio Analysis database were linearly detrended to account for increases in population and aggregate demand. In this analysis, variations in utility revenues were most strongly correlated with annual average PHDI, with 17% of the evaluated water utilities having a Pearson correlation value above 0.5 and an additional 19% having a Pearson correlation value between 0.25 and 0.5. This suggests that 36% of the examined water utilities experience some significant level of vulnerability to drought-related financial risks, as measured by PHDI. The assumption is that the other utilities are either less financially vulnerable to drought based on higher supply-to-demand ratios (which reduces the need to conserve) or employ alternative tools for managing drought-related financial risks (e.g., drought pricing, reserve funds) that are sufficient to reduce their risk to acceptable levels. Index insurance is then used for managing financial risks from severe drought. For a more detailed comparison and integration of the tools used for managing drought-related financial risks for water utilities, see Zeff and Characklis (2013) and Zeff et al. (2014).

Data on PHDI are available for all 344 climate divisions in the United States over a 120-year period from 1895 to 2015. In order to better estimate drought-related financial risks for surface water utilities across the United States (beyond the 372 utilities in the 78 climate divisions considered in earlier research), synthetic PHDI data were produced to extend the PHDI record [based on historic data available from the NOAA Climatic Data Center (2017)]. In creating the synthetic PHDI data, it is important to preserve the spatial and temporal autocorrelation in the observed data across all 344 climate divisions. Autoregressive models that incorporate the spatial autocorrelation of the historic data were used to generate 100,000 years of synthetic PHDI values for each climate division. The spatial autocorrelation matrix was calculated based on the historic PHDI data across all climate divisions and then broken down, using singular value decomposition, to generate a matrix of values ($\mathbf{Z}$) composed of an orthogonal matrix ($\mathbf{U}$), a diagonal singular matrix ($\mathbf{D}$), and a matrix of random deviates from a normal distribution ($\mathbf{X}$) (Onorati et al. 2013), such that

$$\mathbf{Z} = \mathbf{U} * \sqrt{\mathbf{D}} * \mathbf{X} \quad (1)$$

where $\mathbf{U}$ = orthogonal matrix of eigenvectors; and $\mathbf{D}$ = diagonal singular matrix of eigenvalues.

This matrix of random values ($\mathbf{Z}$) that maintains the historic covariance across all 344 climate divisions is a $344 \times 100,000$ matrix composed of random values for each year (100,000) for each utility. These values are then used in the autoregressive models to generate synthetic PHDI values, with models for each climate division determined based on the Akaike information criterion (AIC) and the temporal partial autocorrelation function (Akaike 1974; Shibata 1976). The AIC aids in model selection by selecting a lag time that minimizes model information loss by quantifying trade-offs between the goodness of fit of the model and the total number of parameters. The partial autocorrelation function indicates the influence of each lagged year on overall temporal autocorrelation, controlling for the other years. The autoregressive models for PHDI in each climate division are best fit using either a 1-year lag [AR(1)] (180 climate divisions) or a 2-year lag [AR(2)] model (164 climate divisions) (Fig. 2).

The Thomas–Fiering model has frequently been used in generating synthetic hydrologic data because it maintains the spatial and

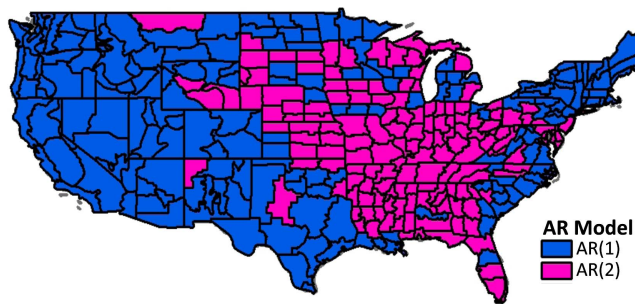


Fig. 2. Number of autoregressive lags determined to be significant for each climate division.

temporal autocorrelation of the historic data (Stedinger and Taylor 1982; Loucks 1992). Synthetic PHDI values are calculated using the Thomas–Fiering Lag-1 Markov model [Eq. (2)] for AR(1) models, and a higher lag modification of the model [Eq. (3)] is used for AR(2) models (Reddy 1997; Singh and Yadava 2003)

$$\text{AR(1) model: } Q_y = \mu + \alpha_1(Q_{y-1} - \mu) + Z_y\sigma\sqrt{1 - \alpha_1^2} \quad (2)$$

$$\begin{aligned} \text{AR(2) model: } Q_y = & \mu + \alpha_1(Q_{y-1} - \mu) \\ & + \alpha_2(Q_{y-2} - \mu) + Z_y\sigma\sqrt{1 - R^2} \end{aligned} \quad (3)$$

where Q_y = average monthly PHDI for the current year; μ = mean of PHDI (based on historic data); α_1 = regression coefficient (for a 1-year lag); Q_{y-1} = PHDI for the previous year; Z_y = value from the Z matrix for the current year; σ = standard deviation of PHDI (based on historic data); α_2 = regression coefficient (for a 2-year lag); Q_{y-2} = PHDI for 2 years ago; and R^2 = coefficient of multiple correlation

$$R^2 = \frac{\rho_1^2 + \rho_2^2 - 2\rho_1^2\rho_2}{1 - \rho_1^2}$$

where ρ_1 = autocorrelation (for a 1-year lag); and ρ_2 = autocorrelation (for a 2-year lag).

Synthetic PHDI data exhibit a similar distribution to historic PHDI, with mean values being nearly identical (historic 0.1501, synthetic 0.1496) and the standard deviation differing only slightly

(historic 2.051, synthetic 1.999) (see Supplemental Data for supporting figure). Spatial autocorrelation of synthetic PHDI values also matches historic observations, fitting a linear model with an R^2 of 0.9987.

Designing Index-Based Insurance Contracts

Even given similar levels of drought, not every surface water utility will be subject to the same level of financial risk because various idiosyncrasies, such as pricing structure, supply-to-demand ratio, or other utility-specific characteristics, will impact each differently. Although information on a utility's specific circumstances in all these areas is not available, a strong correlation between PHDI and utility revenues is assumed to indicate that a utility faces a situation in which it is financially vulnerable to drought. For the set of 372 surface water utilities for which annual operating revenue data are available from Moody's, 17% (63 utilities) and 36% (134 utilities) were identified as having higher levels of drought-related financial risk, as measured by correlations between revenue and PHDI greater than 0.5 and 0.25, respectively (Baum et al. 2018). This subset of high-risk utilities ($R > 0.25$) was further analyzed in an attempt to identify any patterns in location or size that would give rise to additional risk relative to the other utilities without higher correlations between PHDI and revenues, but no patterns were observed.

These results were therefore extrapolated across the United States, such that 36% of the population in each climate division served by surface water utilities was assumed to be representative of the set of utilities considered at risk with respect to drought-related financial losses and to purchase index insurance against severe droughts ($\text{PHDI} \leq -3$). Values for the population served by surface water utilities (as reported in 2017) were obtained from the EPA's Safe Drinking Water Information System (SDWIS) Federal Reports and overlaid with climate division boundaries to determine the population served by publicly operated surface water utilities in each climate division (EPA 2017) (Fig. 3).

Once the synthetic PHDI values and subset of population served by at-risk surface water utilities were determined for each climate division, index-based insurance contracts were created so that the simulation to estimate the distribution of annual payouts for the set of all mutual member utilities nationwide could be developed. Each index-based insurance contract was based on the PHDI and involved the specification of a strike value (k), below which payouts begin, and a slope (m) that reflects the line of best fit between the PHDI and the revenues, which had a slope with units of \$/PHDI unit (Fig. 4). In this case, a contract with a PHDI strike

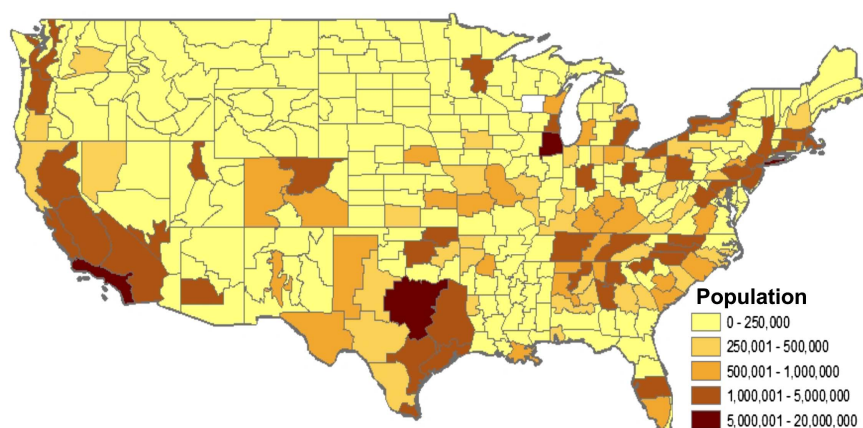


Fig. 3. Current populations served by publicly operated surface water utilities for each of 344 climate divisions.

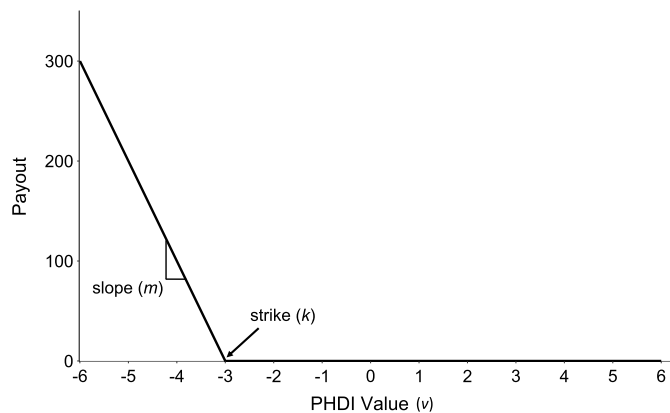


Fig. 4. Generic structure of PHDI-indexed insurance payout with a strike of -3 .

value of -3 (severe drought) was established for all 344 climate divisions. The slope for each contract varied based on the relationship between PHDI and revenues existing in that climate division, as determined by a generalized relationship between PHDI and revenues scaled to the population served by at-risk water utilities in that specific climate division.

The generalized population-weighted relationship between PHDI and revenues was determined based on the median slope per capita across the set of 134 at-risk utilities, as established from earlier work (Baum et al. 2018). The payout slope per capita for each of these utilities was determined by dividing each utility's payout slope (i.e., line of best fit between revenues and PHDI) by its population served. The median slope per capita across the 134 utilities was 0.005 \$/PHDI unit, meaning that there was a 0.005 increase in the slope of the payouts for each additional person in the climate division. This value was used with the subset of the population served by at-risk utilities in each climate division to calculate the payout slopes for each of the 344 climate divisions of the contiguous United States (see Supplemental Data for each of the 344 slopes). For example, a climate division with a population of 100,000 served by at-risk utilities would have a payout line slope of 500 (\$/PHDI unit), whereas a climate division with a population of 1,000,000 served by at-risk utilities would have a contract with a payout line slope of 5,000 (\$/PHDI unit).

Once the index-based insurance contracts were designed for the at-risk utilities in each climate division, the payouts could be

determined over the synthetic PHDI data set of 100,000 years. At each annual time step, the aggregate annual payout was determined based on the annual average of monthly PHDI values in each climate division (v), the strike value (k) of -3 , and the payout slope (m), a unique payout value for each climate division (i), such that

$$\text{Aggregate annual payout} = \sum_{i=1}^{344} [m_i * \text{MAX}((k - v_i), 0)] \quad (4)$$

Simulating payouts over the synthetic period of 100,000 years yielded an aggregate annual expected payout of \$20.5M across all 344 climate divisions. Although payouts were less than this expected level 73% of the time, they were more than twice this level in 16% of years, and more than 8 times the expected payout 1% of the time (Fig. 5). These low-probability, high-consequence years with very high payouts make a strong case for employing reinsurance, as the mutual's required reserves would need to be maintained at a very high level to manage these events in a way consistent with regulatory requirements (up to the 99.5% VaR). There may, therefore, be significant potential for the reserve size to be reduced with reinsurance, lowering the mutual's cost of capital and presumably reducing the costs for its insured utilities.

Pricing Index Insurance Contracts

The total price for insuring the populations served by at-risk surface water utilities across all 344 climate divisions was determined by the expected payouts to insured parties and an additional loading factor, which accounts for a number of factors, including the costs of administration, research, and marketing

$$\text{Premium} = E[\text{payout}] + \text{loading} \quad (5)$$

The loading for the mutual was assumed to be a constant fraction of the expected payout, in this case 25%, based on average expense ratios of mutuals (Hartwig and Alldredge 2018). Based on the aggregate annual payout value of \$20.5M, the loading was \$5.13M.

The mutual achieves the required level of reserves by issuing a 30-year bond in an amount reflective of the 99.5% VaR (\$212M), for which it makes constant annual payments. Based on an average of the range of investment-grade bonds (rated Baa to AAA) over the period 2008–2018, the bond coupon rate is 4.92% (MSRB 2019; FRED 2019). Using this value, the amortized annual bond

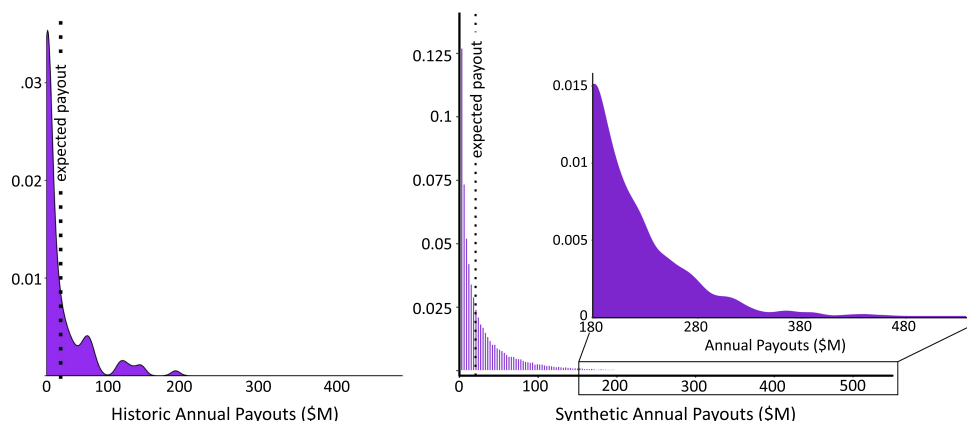


Fig. 5. Distribution of the historic and synthetic annual payouts for the risk pool and the highest 1% of annual payouts.

payment for the mutual is \$13.6M (on the \$212M bond). The cost of capital can be substantial, and in this case, with a required reserve level more than 10 times the expected annual payout (\$20.5M), the mutual will have to maintain a significant fraction of the reserves that will very rarely be used. Transferring some of this more severe tail risk is likely to be cost effective.

Designing Reinsurance Contracts

In dealing with tail risks, mutual insurers often purchase “excess of loss” reinsurance contracts, in which the mutual is responsible for risks up to a specific level, the “attachment point,” above which the reinsurer is responsible for risks until some agreed-upon limit. The remainder of the risk above the limit is assumed by the mutual. In some cases, a reinsurer may cover the majority, but not all, of the risks between the attachment point and limit, leaving the buyer of the reinsurance (in this case, the mutual) responsible for some portion of losses in this “layer.” This is typically insisted on by the reinsurer as a means of combating moral hazard concerns and ensuring that the party taking out the reinsurance is incentivized to take reasonable measures to prevent losses. However, because the reinsurance contracts considered here are index based, moral hazard concerns are minimal (NAIC 2018), and the contract is designed such that all losses between the attachment point and the limit are covered by the reinsurer.

Reinsurance contracts are evaluated across a range of attachment points and limits, with attachment points ranging from \$30M to \$200M (in \$10M increments), whereas limits range from \$40M to \$212M (in \$10M increments up to \$200M and then jumping to \$212M). All combinations of attachment point and limit are assessed across different layer sizes, ranging from \$10M layers (e.g., \$60M attachment point and \$70M limit) to \$182M layers (\$30M attachment point and \$212M limit). In total, 171 different reinsurance contract layers are assessed to identify tradeoffs across these combinations through determining the net cost of risk management, which will be defined in the next section.

Pricing Reinsurance Contracts

Reinsurance contract prices can be more difficult to estimate given a relative dearth of historical data, creating greater uncertainty in modeling these higher-risk events (Qiu et al. 2012). Because there are fewer data and greater variability in the distribution of the risks managed via reinsurance relative to most insurance contracts, loading values for reinsurance tend to be higher (Froot 2001; Cummins et al. 2009). Although theoretical models are frequently used to price insurance (Hardy 2006), reinsurance is often subject to less generalizable pricing approaches that involve a mix of both rigorous actuarial analysis and negotiations between the counterparties (Fu and Khury 2010; Lane and Mahul 2008).

This work focuses on an increasingly common type of reinsurance, insurance linked securities (ILSs), in which the reinsured risk is fully securitized by an investor (McKinsey & Company 2013; Chang, C. W., J. S. K. Chang, and M. Y. Yu. 2014. “Embracing the perfect storm: A theory of optimal catastrophe risk financing,” working paper, California State Univ., Fullerton). This securitization indicates that the value of the reinsurance is held in a liquid form and available immediately when needed, an important consideration for a mutual. The specific ILSs of interest in this work are collateralized reinsurance contracts, which are structured similarly to publicly traded catastrophe bonds (another type of ILS) but privately arranged. Because catastrophe bonds are publicly traded, their pricing history is available (Artemis Deal Directory 2018), and these data have been used to develop empirical pricing models.

These models have shown that important components in determining catastrophe bond prices include, primarily, the expected covered losses, but other factors such as peril type (e.g., wind and flooding), peril territory, market cycle, trigger type, and size can also be significant influences (Braun 2016; Galeotti et al. 2013; Chang, C. W., J. S. K. Chang, and M. Y. Yu. 2014. “Embracing the perfect storm: A theory of optimal catastrophe risk financing,” working paper, California State Univ., Fullerton; Gatumel and Guegan 2008; Dickemann 2009). Although there is agreement that all these characteristics have some effect on catastrophe bond prices, the magnitude of the effects varies.

Because there is no established method to price drought-related reinsurance contracts, the best-performing catastrophe bond pricing model (Braun 2016), from among five alternative models studied in the literature, was used in this work. This model, which Braun developed based on catastrophe bond price data from 1997 to 2012, is a multivariate linear econometric model, in which coefficients for six variables determined to significantly affect pricing were identified (expected loss, location, diversification of the sponsor’s portfolio, market conditions, credit rating, and the price of comparable investments):

$$P_{Re} = 221.04EL + 175.08PEAK - 103.58SW + 161.85ROLX - 159.76IG + 26.57BBSPP \quad (6)$$

where EL = expected layer payout (in percentage points); PEAK = peak territory designation (1 if yes; 0 if no); SW = SwissRe sponsored deal (1 if yes; 0 if no); ROLX = synthetic rate on line index (Lane Financial LLC index); IG = investment grade rating (1 if yes; 0 if no); and BBSPP = BB corporate bond spread (in percentage points).

The other five models assessed were functions of either solely expected loss, expected and conditional expected loss, or expected loss and the market price of risk. The Braun model resulted in a lower out-of-sample root-mean-square error than the other models by at least 5.4% and at most 17.4% (Braun 2016). In the Braun model, the expected loss (EL) had the greatest impact on the reinsurance premium. Intuitively, catastrophe bonds with higher expected losses, and thereby higher risk, will be more expensive. The second-largest coefficient was associated with the peak territory (PEAK), a binary variable of 1 or 0, that reflects the higher level of catastrophe risk capital concentrated in the United States. Because these US-based risks do not aid in geographically diversifying the portfolio of risks and thereby contribute to greater correlation among geographically colocated risks, the peak territory designation was set to 1. The third term (SW) is another binary variable that relates to the diversification of the reinsurer, specifically SwissRe, which issues the majority of catastrophe bonds, and because of its high level of diversification tends to be in a position to offer lower reinsurance premiums. This work assumed that the reinsurer will be similarly well diversified, and so the value of SW was set to 1. The rate on line index (ROLX) is a synthetic index compiled by Lane Financial Company that accounts for the changes in the market cycles for reinsurance. The average for this value over the historic period of catastrophe bond pricing for which the ROLX is available (1997–2012) is 0.95, which was used for this work. The investment grade rating (IG) allows for higher-rated bonds to reflect lower risk through lower reinsurance premiums. Because the majority of catastrophe bonds are rated BB (noninvestment grade), this value was set to 0. For the final term, the BB corporate bond spread relative to the spot Treasury curve (BBSPP), the average spread over the last 20 years is 3.5%, so this value was used. Therefore, when applying the Braun model across all possible

Table 1. Reinsurance contract prices and loadings for various attachment points and a limit of \$212M (99.5% VaR) and associated annual amortized debt service costs

Attachment point (in \$M)	Debt service payment (in \$M)	Expected layer payout (in \$M)	Braun contract premium (in \$M)	Braun contract loading (%)
30	1.93	9.37	26.50	183
40	2.58	7.54	22.13	193
50	3.22	6.1	18.62	205
60	3.87	4.95	15.76	219
70	4.51	4.01	13.38	233
80	5.16	3.25	11.38	250
90	5.80	2.64	9.70	268
100	6.45	2.13	8.25	288
110	7.09	1.71	7.01	311
120	7.74	1.36	5.92	336
130	8.38	1.07	4.96	364
140	9.02	0.83	4.11	395
150	9.67	0.63	3.36	431
160	10.31	0.47	2.68	470
170	10.96	0.34	2.06	514
180	11.60	0.23	1.50	565
190	12.25	0.14	0.99	621
200	12.89	0.07	0.51	680

reinsurance layers to this work, the final pricing model to determine the reinsurance premium is

$$P_{Re} = 221.04EL + 304.97 \quad (7)$$

When calculating the reinsurance contract premiums for attachment points from \$30M to \$200M and a limit of \$212M, the premiums ranged from \$0.5M to \$26.5M and the loadings from 183% to 680% (Table 1). These high loading values, which are in line with observed values, indicate the cost of insuring against low-probability, high-risk events, which accounts for the costs of maintaining high levels of capital.

Because this work is concerned only with parametric contracts, which represented 18% of catastrophe bonds from Braun's analysis, the Braun pricing model was tested over this subset of data. For the 116 parametric contracts, model prices generated with the Braun model aligned strongly with observed prices from the Artemis Deal Directory for parametric contracts, with a Pearson correlation of 0.84 (Fig. 6).

The accuracy of the Braun pricing model on parametric contracts suggests that it reasonably represents the pricing for reinsurance contracts for this PHDI based insurance. However, because this reinsurance contract pricing is not perfect, a sensitivity analysis on the modeled prices was conducted.

Determining Risk Exposure with Reinsurance

The mutual establishes the reserves required for its operations by issuing a 30-year bond in an amount that allows it to meet regulatory requirements. Without reinsurance, reserves are the only means of covering claims by mutual members, and the bond will be in the amount of \$212M, equal to the regulatory reserve requirement (99.5% VaR). In the case where reinsurance is employed, the bond size is commensurately reduced, depending on the level of reinsurance coverage, with the total combined coverage of reserves and reinsurance amounting to \$212M of coverage (i.e., if a reinsurance contract covers losses between \$150M and \$212M, the bond used to establish reserves will be issued for \$150M) (Table 1).

Accounting for the variable levels of debt service and reinsurance loading costs, the total cost for the insured members can be represented as

$$\text{Net Cost} = \text{Mutual Loading} + \text{Reinsurance Loading} + \text{Debt Service} \quad (8)$$

Each year, the mutual's reserves may rise or fall, depending on the total payouts in a particular year. If no contracts are triggered and there are no payouts, the reserves increase by an amount equal to the premium received less the administrative costs. The reserves will also increase if lower-than-expected payouts are disbursed. A credit rating agency, A. M. Best, found that 75% of rated mutual insurers maintained capital in excess of 250% of the regulatory requirement in order to ensure long-run solvency (Best 2015; NAIC 2015; Laux and Muermann 2009; Gonulal 2012; Harrington and Niehaus 2002; Pain et al. 2016). Therefore, reserves up to the 99.5% VaR regulatory level are maintained in a very liquid form and, in this work, earn only 0.53% (consistent with 6-month Treasury bills) (FRED 2019), whereas any reserves above this level are maintained in low-risk investments, earning 4.35%, consistent with AAA-rated corporate bonds (FRED 2019). These assumptions apply to reserves in scenarios involving reinsurance and those without.

In order to account for fluctuations in reserves over time and determine how they impact Net Cost for mutual members, a 10-year time series was assessed. Each year, reserves must be at least as high as the 99.5% VaR (\$212M), so additional capital or reinsurance coverage may be required after years with high payouts. If reserves fall below the 99.5% VaR, the mutual insuring with reinsurance purchases a new reinsurance contract, with an attachment point equal to the current level of reserves and a limit equal to the 99.5% VaR (\$212M). The reinsurance contract that results in the lowest average Net Cost for any individual year (see results in next section) is used as the reinsurance contract attachment point at the beginning of each 10-year period. Subsequent years will result in reinsurance contracts with different attachment points and thereby different reinsurance loadings (which affect Net Cost), based on the level of reserves. If, over time, no payouts are made and reserves build up to the 99.5% VaR, then no reinsurance will

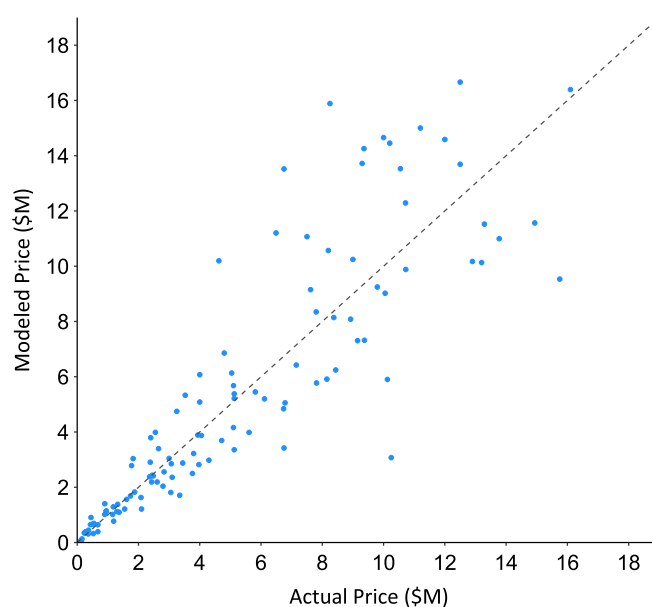


Fig. 6. Modeled catastrophe bond prices based on the Braun pricing model compared to historic actual prices.

need to be purchased. In the case of the mutual insuring without reinsurance, more debt is issued in any year with a deficit relative to the 99.5% VaR at year end. This additional debt is issued in the same manner as the initial debt, over a 30-year time span and a coupon rate of 4.92%. The new amortized bond payment is then added to the Net Cost in each year until the debt service payments are fulfilled after 30 years.

Results

The Net Cost for mutual members, as well as the mutual's reserve levels (with and without reinsurance), were compared for 171 reinsurance contract structures. Across all of these contract structures, the reinsurance upper limit of \$212M always resulted in the lowest average Net Cost, as the loading nominally decreased across each higher layer due to the lower probability of payouts, whereas the debt service cost increased linearly. Therefore, this section will focus only on the 18 unique contracts with that limit.

Average Net Cost for Mutual Members

Before assessing the 18 scenarios, the components of Net Cost, including debt service and reinsurance loading, are examined to identify the tradeoffs between them. In general, rising debt service costs (i.e., larger reserves) correspond to higher reinsurance attachment points and lower reinsurance loading, whereas lower attachment points lead to reduced debt service costs and higher reinsurance loading (Fig. 7). Reinsurance is the primary risk management tool when the contract has a low attachment point such as \$30M, and the reinsurance loading in this case is quite high (\$17.1M), and this allows for smaller reserves, which lowers debt service costs to \$1.9M. Debt service costs rise when larger reserves are paired with reinsurance contracts with high attachment points, even as reinsurance loading falls. All of these combinations of insurance and reinsurance offer coverage of losses up to at least the 99.5% VaR (\$212M), but there is a tradeoff between debt service costs and reinsurance loading costs as the relative roles of reserves and reinsurance shift. The sensitivity analysis of the Braun pricing model describes differences in reinsurance loadings of up to \$2.6M, and this disparity is largest for situations in which the reinsurance attachment point is very low, situations unlikely to be the most cost effective (see the next section).

When calculating a mutual member's Net Cost (which includes the mutual's loading, as well as reinsurance loading and debt service), the use of reinsurance with an attachment point above \$75M resulted in lower Net Cost compared to sole reliance on reserves (no reinsurance) (Fig. 8). The lowest average Net Cost across all reinsurance contracts was \$17.4M, which was \$1.4M lower than the Net Cost of insuring without reinsurance (\$18.8M), resulting from insuring with a reinsurance contract with an attachment point of \$130M. Employing reinsurance contracts with attachment points below \$80M resulted in a higher Net Cost compared to using reserves alone (i.e., no reinsurance).

When considering the sensitivity bounds of the analysis, a 10% increase in the reinsurance premiums resulted in a lowest average Net Cost of \$17.8M, based on a reinsurance contract attaching at \$140M. For this set of higher priced reinsurance contracts, attachment points above \$95M resulted in a lower average Net Cost when compared to insuring without reinsurance. For a 10% decrease in reinsurance premiums, a reinsurance contract attaching at \$110M resulted in the lowest average Net Cost of \$16.8M, with all contracts attaching above \$55M resulting in lower average Net Cost compared to insuring without reinsurance.

Total Net Cost for Mutual Members over 10 Years

Although considering the average of individual years is useful, using a longer time horizon to evaluate strategies can also be informative because it allows for a better understanding of various path-dependent considerations, such as the rising and falling of reserve levels. A 10-year period was used as representative for planning purposes, and all analyses over this period involved starting the period with the least expensive scenario, which involved reinsurance with an attachment point of \$130M. Over 10,000 trials of simulated 10-year periods, the average Net Cost was \$212.3M when considering a baseline scenario involving no reinsurance and \$177.1M for the mutual operating with reinsurance, a 16.6% decrease (Fig. 9). A mutual operating without reinsurance may lead to the highest 1% of Net Cost over 10 years of \$383M because subsequent years of high payouts result in multiple new debt issuances to maintain sufficient reserves to meet the mutual's 99.5% VaR (\$212M) objective. For the mutual operating with reinsurance, the highest 1% of Net Cost over 10 years was only \$282M, which is a 26% reduction. Although reinsurance contracts with low attachment points may be expensive when reserves drop to low levels,

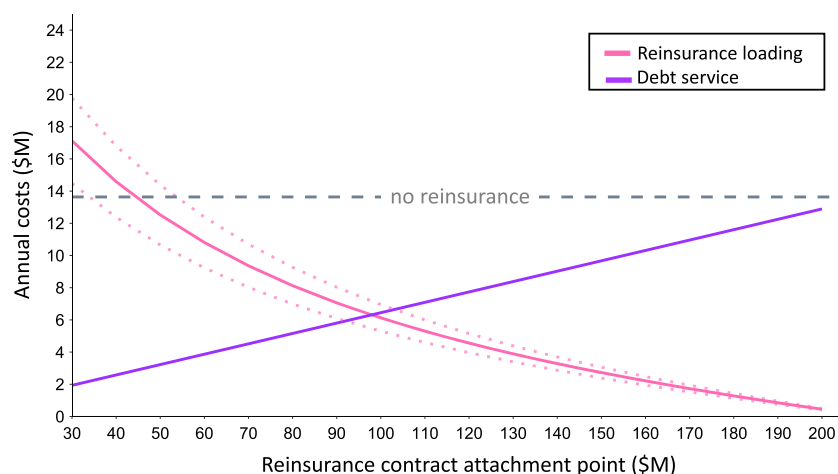


Fig. 7. Tradeoff between annual reinsurance loading and debt service costs for insuring with reinsurance at various attachment points compared to insuring without reinsurance. The lower and upper bounds (dashed lines) around reinsurance loading indicate $\pm 10\%$ sensitivity limits.

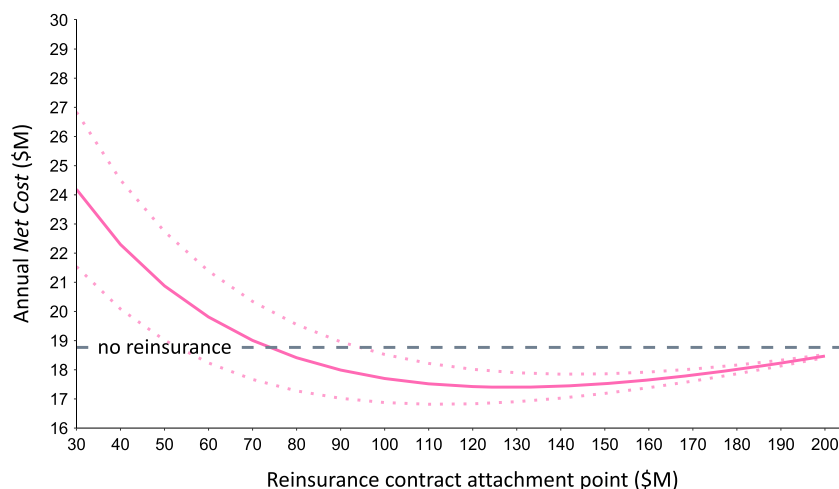


Fig. 8. Net cost (annual) for insuring with reinsurance at various attachment points compared to insuring without reinsurance. The lower and upper bounds (dashed lines) indicate $\pm 10\%$ sensitivity limits.

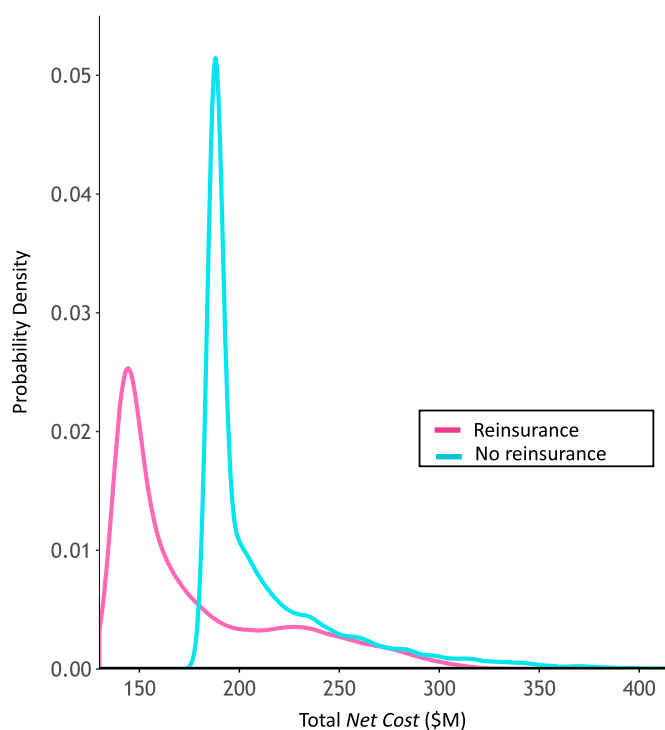


Fig. 9. Net cost across 10 years for insuring with and without reinsurance, beginning at a \$130M attachment point when insuring with reinsurance.

they are significantly less expensive than multiple years of additional debt service payments.

Purchasing reinsurance contracts, as opposed to issuing new debt, as a means of maintaining protection at the 99.5% VaR level resulted in lower Net Cost, whereas operating without reinsurance leads to higher reserve buildups over the 10-year period. Average reserves at the end of the 10-year period were \$175M and \$316M when insuring with and without reinsurance, respectively (Fig. 10).

The higher reserve levels are an important consideration because they require incurring more debt, and outstanding debt at the end of each 10-year period can be substantial for the mutual insuring without reinsurance. As more debt is issued to increase reserves,

this results in larger average reserve levels for the mutual insuring without reinsurance over 10 years, given that reserves are always at least \$212M. The debt service costs on these larger reserves are, of course, somewhat offset by the higher capital gains (4.35% for reserves above \$212M) relative to a situation involving a mutual insuring without reinsurance (earning 0.53% for reserves up to \$212M). In order to account for both the outstanding debt, which would decrease reserves, and the projected capital gains on reserves, which would increase them, a new metric, Net Reserves, was computed for the mutual both without and with reinsurance and is presented as an aggregate 10-year amount. Although the present value of outstanding debt is accounted for by discounting future debt service payments to the end of year 10, projected capital gains are dependent on fluctuating reserve levels over the 30 years. Therefore, historic average capital gains, which follow a linear growth pattern, were used as an appropriate metric for both the mutual with and without reinsurance and used then to project average capital gains over the 30-year bond period and discounted to the present value (PV)

$$\begin{aligned} \text{Net Reserves} = & \text{Reserves} - \text{PV}(\text{Outstanding Debt}) \\ & + \text{PV}(\text{Project Average Capital Gains}) \quad (9) \end{aligned}$$

The average Net Reserves after 10 years were \$214M and \$197M, respectively, for insuring without and with reinsurance (Fig. 11). Although the Net Reserves were, on average, 8% higher for the mutual insuring without reinsurance, the mutual insuring with reinsurance had a lower probability of insolvency (i.e., reserves falling below zero) over the 10-year period, with only 0.17% of the Net Reserves falling below zero, whereas insolvency occurred in 9.5% of the 10-year simulations for the mutual insuring without reinsurance. Without reinsurance, the mutual will often build up reserves to a higher level, in this case a maximum of \$445M. In the case of the mutual insuring with reinsurance, reserves did not rise higher than \$376M over the 10,000 simulations run. However, these additional projected average capital gains are most likely outweighed by the significantly higher risk of insolvency when insuring without reinsurance.

Discussion

When considering the concept of a mutual for managing water utilities' drought-related financial risks, using reinsurance to transfer

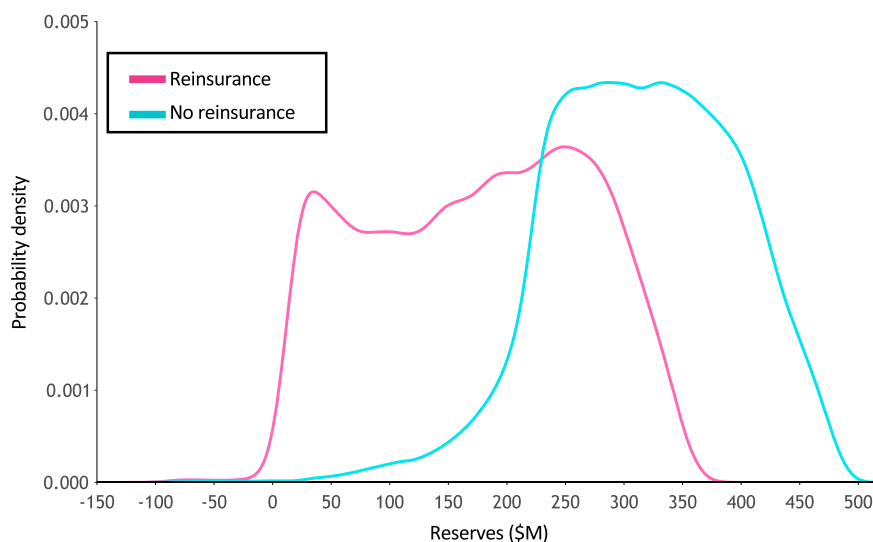


Fig. 10. Reserves after 10 years for insuring with and without reinsurance, beginning at a \$130M attachment point when insuring with reinsurance.

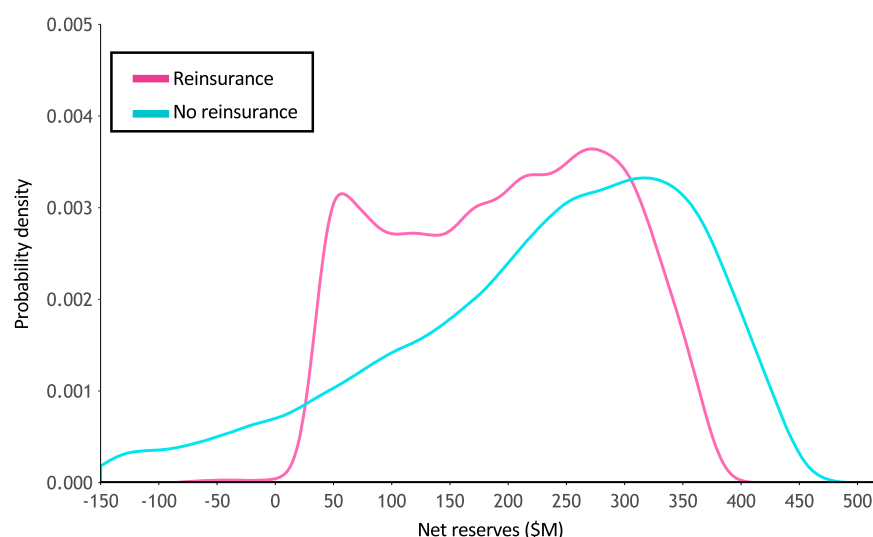


Fig. 11. Net reserves after 10 years for insuring with and without reinsurance, beginning at a \$130M attachment point when insuring with reinsurance.

tail risks can help reduce the Net Cost to members and reduce variation in Net Reserves over time. Although water utility managers have not yet investigated the use of financial instruments widely, their development and use have been mentioned in industry literature in recent years (Beecher and Chesnutt 2012; Hughes et al. 2014; AWE 2014). There is also evidence that other public sector actors are beginning to consider reinsurance and other forms of financial risk transfer, given that the Federal Emergency Management Agency recently purchased its first catastrophe bond in August 2018 to better manage risks for the National Flood Insurance Program. As the use of financial instruments continues to grow, understanding the degree to which reinsurance might play a role in managing financial risk related to hydrologic variability, within the water utility sector and beyond, may have increasing value.

In this work, a mutual serves a large number of member utilities with mostly uncorrelated risks, so pooling of these risks has some advantages in terms of reducing the highest potential losses in any

given year. However, if mutual members were heavily concentrated in a specific geographic region, say Southern California, the correlation between the financial risks of the members could dilute the value of risk pooling and give rise to concerns over even larger payouts in drought years. Under such circumstances, more tail risk would likely be shifted via reinsurance, making it even more valuable in terms of reducing members' Net Cost. In this case, the cost-reducing potential of reinsurance would, of course, be offset by the higher premiums resulting from a more highly correlated pool of risks.

With respect to other areas that would have improved this analysis, additional data from water utilities on factors affecting their financial risk from drought, such as pricing schedule and supply-to-demand ratio, would likely reduce the basis risk of index-based contracts and improve their effectiveness. In this work, the financial exposure of 372 geographically dispersed water utilities spread across 78 US climate divisions were extrapolated to analyze the financial risk exposure of an even greater spread of water utilities

across all 344 US climate divisions. Data from additional water utilities and information regarding specific characteristics of each water utility would further help to improve assumptions regarding the subset of utilities across the United States at risk and thereby the results of the analysis.

The Braun reinsurance pricing model indicates a high level of accuracy for pricing index-based reinsurance contracts, and varying prices across a sensitivity range suggest that reinsurance is likely to reduce costs under many circumstances. Traditional reinsurance prices were driven down by competition from newly developed catastrophe bonds, and continued growth will likely drive prices even lower in the future (Chang, C. W., J. S. K. Chang, and M. Y. Yu. 2014. "Embracing the perfect storm: A theory of optimal catastrophe risk financing," working paper, California State Univ., Fullerton). Should this occur, reinsurance is likely to appear even more beneficial than it does in this analysis. Other assumptions affect both the scenarios of insuring with and without reinsurance, including the bond coupon rate of 4.92% for the 30-year bond issued to raise the initial reserves, the earnings of 0.53% on the reserves up to the 99.5% VaR, and earnings of 4.35% for low-risk investments for any reserves beyond the 99.5% VaR. Variation in any one of these rates will likely similarly affect the other rates, resulting in similar findings.

Conclusion

With predicted increases in the frequency and severity of extreme weather events, it will be important to continue searching for improved solutions to manage the financial risks of drought. Water utilities, in particular, should be cognizant of a range of tools and strategies available to them as part of their risk management practices, especially given that the financial consequences of drought are being more carefully scrutinized by credit rating agencies, increasing the threat of higher interest rates on debt (Chapman and Breeding 2016). Reducing these risks in a cost-effective manner requires identifying new risk management strategies that will become increasingly important in the future.

This work analyzes the strategy of reinsurance as an additional strategy to use with mutual insurance that can result in an annual net cost reduction of 7.5% and a 10-year net cost reduction of 17%. Additionally, the probability of insolvency over a 10-year period is only 0.17% when using reinsurance compared to 9.5% without it. As newer risk management strategies are analyzed, they will need to be assessed alongside the existing risk management practices of water utilities in order to determine the most cost-effective practices. Water utility personnel, financial institutions, and urban planners, all of whom will need to consider the risk of financial losses from drought, may find useful insights in this work, as may those that create and rate the debt vehicles that fund investments in water utility infrastructure.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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Supplemental Data

Fig. S1 and Table S1 are available online in the ASCE Library (www.ascelibrary.org).

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